

# TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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June 18, 2026 · v5.1 (rev 180)

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading document 5 (Red Team).

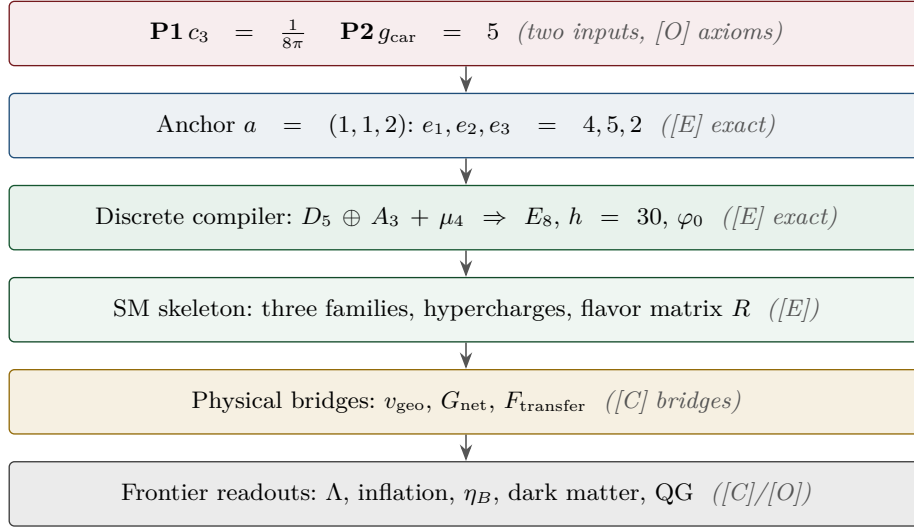
## TFPT status at a glance — read this card the same way in every document

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed (**[E]**); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status                  |
|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive <b>[O]</b>    |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | <b>[E]</b> / <b>[C]</b> |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | <b>[C]</b>              |

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}$ ,  $G_{\text{metric}}$ ,  $F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**[E]** exact / machine-proven**[C]** conditional**[O]** open / axiom**[X]** kill test**What this note is**

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by `[verification/redteam/rt_A_e8net.py]` ... `[verification/redteam/rt_E_vgeo.py]` and `[verification/redteam/run_redteam.py]`; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

| Target | Minimal claim                                            | Status                     | Residual risk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|----------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A      | seam–Calderón measure = $(E8)_1$ net                     | <b>reduced, not closed</b> | <i>one</i> statement: boundary-net holomorphy + $c=8$ ( $\Leftrightarrow$ the index-4 inclusion); $E_8$ and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Since reduced to <i>zero new</i> open content: the free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the already-open $A_2$ (net assembly) + GATE.QGEO (v160–v165). The three-residual form below is the historical development. |
| B      | $g_{\text{car}} = 5$ forced by the Pascal rule           | <b>survives (narrowed)</b> | boundary derivation of half-spinor exhaustion (degree-2 truncation)                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C      | $k = c_3/2 = \frac{1}{16\pi}$ , $S/A = \frac{1}{4}$      | <b>survives</b>            | absolute $1/G$ (UV-sensitive) is the one anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| D      | ratios+products $\Rightarrow$ one scale $v_{\text{geo}}$ | <b>survives (narrowed)</b> | CP phases ( $\delta_{\text{CKM/PMNS}}$ ) not covered by $v_{\text{geo}}$                                                                                                                                                                                                                                                                                                                                                                                                                             |
| E      | one scale $v_{\text{geo}}$ carries the theory            | <b>survives (narrowed)</b> | EW/reheating/leptogenesis scales + seam=Planck identification                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Three outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), and **reduced, not closed** (the conservative wording is the correct one). A fourth, **broken**, is reserved for an actual failure; none occurred.

## Target A — the $(E8)_1$ boundary-net identification

**Minimal claim.** The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the  $(E8)_1$  lattice net, carrier subnet  $(D5)_1 \times (A3)_1$ . [verification/redteam/rt\_A\_e8net.py]

**Logical chain (established).** The Sugawara level-1 central charges are  $c(E8)_1 = \frac{248}{31} = 8$ ,  $c(D5)_1 = 5 = g_{\text{car}}$ ,  $c(A3)_1 = 3 = N_{\text{fam}}$ , so the conformal-embedding criterion  $c_{\text{coset}} = c(E8) - c(D5) - c(A3) = 0$  holds [E]. This is *compatibility*.

### Where it breaks: central charge underdetermines the net

Counterexample:  $(D8)_1 = SO(16)_1$  also has  $c = \frac{120}{15} = 8$ , but is *not* holomorphic (four primaries  $1, v, s, c$ ) — a distinct  $c = 8$  net. Hence equal central charge is *necessary, not sufficient*:  $(E8)_1$  is the unique *holomorphic*  $c = 8$  net (even unimodular rank-8 lattice =  $E8$ ), so **holomorphy (single vacuum sector,  $\mu$ -index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the  $c = 8$  Narain family is positive-dimensional. The constructive map seam–Calderón kernel  $\rightarrow (E8)_1$  and the uniqueness of bulk reconstruction are not established; the gap  $\Delta_{\text{eff}} > 0$  *supports* tightness (clustering) but does not prove it.

**Verdict.** Keep “reduced to a rigorous boundary-net identification problem”; do *not* write “metric sector closed”. The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

## Target B — carrier rank / the Pascal condition

**Minimal formal claim.**  $2^g = g^2 + g + 2$  has the unique solution  $g = 5$  (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion  $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ? [verification/redteam/rt\_B\_pascal.py]

### Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it:  $2^g = g^2 + g + c$  gives  $\{5\}$  only at  $c = 2$ ; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with  $2^{g-1} = \sum_{k \leq K} \binom{g}{k}$  one finds  $K=0 \rightarrow g=1$ ,  $K=1 \rightarrow 3$ ,  $K=2 \rightarrow 5$ ,  $K=3 \rightarrow 7$ , i.e.  $g = 2K + 1$ . So choosing  $K = 2$  (to reach  $g_{\text{car}} = 5$ ) is a physical postulate. It is corroborated — the  $D5$  half-spinor  $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$  and the family closure  $N_{\text{fam}} = (2^{g-1} - 1)/g$ ,  $g + N_{\text{fam}} = 8$  both pick 5 — but each is itself a postulate, not a theorem.

**Verdict.** Theorem A’s arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

## Target C — $k = c_3/2$ and the seam-area coefficient

**Minimal claim.** In reduced units  $k = c_3/2 = \frac{1}{16\pi}$  and Fursaev–Solodukhin gives  $S/A = \frac{1}{4}$ . [verification/redteam/rt\_C\_kc3.py]

**The dimensional firewall holds**

The only legal chain is

$$k_{\text{red}} = \frac{c_3}{2} = \frac{1}{16\pi} \text{ (dimensionless)}, \quad k_{\text{phys}} = \frac{k_{\text{red}}}{G}, \quad S = 4\pi k_{\text{phys}} A_{\text{phys}} = \frac{A_{\text{phys}}}{4G}.$$

The forbidden form  $k_{\text{phys}} = c_3/2$  is dimensionally false (legal only at  $G = 1$ ). A scan of **v67/v68/v73** finds every  $c_3/2$  used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall's teeth are verified against a synthetic violation. Internally consistent with  $1/G$  being UV-sensitive (Sakharov/Connes, **v68**).

**Verdict.** Safe as worded (reduced coefficient). Status **[E]** (structure) + **[O]** for the residual. Residual: the absolute  $4D$  Newton scale  $1/G$  ( $\propto \Lambda^2 f_2$ ) stays the one dimensionful anchor; nothing *dimensionless* is open.

**Target D** —  $U_{\text{point}} \rightarrow v_{\text{geo}}$ 

**Minimal claim.** Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale  $v_{\text{geo}}$ . [[verification/redteam/rt\\_D\\_upoint.py](#)]

**Where it narrows: the bijection needs explicit hypotheses**

For positive reals, (ratios, product)  $\Leftrightarrow$  (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational  $c$ , odd size  $3 = N_{\text{fam}}$ ). But it *fails* off-assumption: even sector size leaves a sign ambiguity ( $\{2, 3\}$  vs  $\{-2, -3\}$  share ratio  $3/2$  and product 6); complex amplitudes leave an  $n$ -th-root branch ambiguity ( $\{1, 1, 1\}, \{\omega, \omega, \omega\}, \{\omega^2, \omega^2, \omega^2\}$  share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases  $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$  are complex and are *not* folded into  $v_{\text{geo}}$ .

**Verdict.** Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in **v75** and must be named. Status **[E]** (reduction) + **[O]** ( $v_{\text{geo}}$ ). Residual: the CP-phase sector.

**Target E** —  $v_{\text{geo}}$  as the unique dimensional floor

**Minimal claim.** No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit  $v_{\text{geo}}$ . [[verification/redteam/rt\\_E\\_vgeo.py](#)]

**Single-scale audit: exact for the certified tiers, conditional beyond**

The closed compiler + protected IR readouts all reduce to (pure number)  $\times v_{\text{geo}}$ :  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ ,  $f_a = (c_3/|\mu_4|)^{7/2} \bar{M}_{\text{Pl}}$ , masses  $= (\pi/\sqrt{2}) c_f \varphi_0^k v_{\text{geo}}$ , and  $1/G$  shares the same anchor (**v68/v75**). The audit *surfaces* candidates that are not pure reductions: the seam cutoff  $\Lambda$  (only an *identification* seam=Planck collapses it to  $v_{\text{geo}}$ ), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier **[C]/[O]** inputs that must not be silently absorbed into  $v_{\text{geo}}$ .

**Verdict.**  $v_{\text{geo}}$  is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status **[E]** (ratios) + **[O]** (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

**Outcome**

**What the red team changed.** No target broke. The hardest gate (A) is confirmed *open* with the conservative wording; three targets (B, D, E) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states

where each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

**Follow-up — the verdicts were then used as a worklist [E]**  
 ([verification/v83\_e8net\_holomorphic\_uniqueness.py])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows GATE.METRIC.04 and CAR.PASCAL.01.

**Target A, residual 1 — closed [E].** At level 1 the number of primaries equals  $\det(\text{Cartan}) = |\text{fundamental group}|$ :  $A_3=4$ ,  $D_5=4$ ,  $D_8=4$ ,  $\mathbf{E}_8=1$ . So holomorphy (single vacuum sector,  $\mu$ -index 1) is *necessary* — it excludes the same- $c$  rival  $(D_8)_1 = SO(16)_1$  ( $c = 120/15 = 8$  but four primaries  $1, v, s, c$ ) — and *sufficient*: a holomorphic  $c = 8$  chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one,  $E_8$ , by the Minkowski–Siegel mass formula ( $\text{mass} = 1/|W(E_8)| = 1/696729600$ , and  $|\text{Aut}(E_8)|$  saturates it). The  $(D_5)_1 \times (A_3)_1$  embedding  $c = 5 + 3 = 8$  stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with  $c = 8$  (then  $E_8$  is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy +  $c = 8$  ( $\Delta_{\text{eff}} > 0$  supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [C]/[O].

**Target B — reduced [E].** The “ $K = 2$  Pascal truncation” is not free:  $K = (g - 1)/2$  is the Pascal-row midpoint  $\sum_{k \leq (g-1)/2} \binom{g}{k} = 2^{g-1}$  (odd  $g$ ), i.e. the half-spinor split; the carrier is the even Clifford half-spinor  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$ . So B’s residual reduces to the single standard input “carrier = half-spinor of  $\text{Spin}(10)$ ” (the Lean arithmetic core stays [E]).

**Target B — a backward certificate ([verification/v219\_icosahedral\_mckay.py]).** The choice  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$  that B attacks is, *given* the  $E_8$  closure, the icosahedron: by the McKay correspondence  $E_8$  is the exceptional top of the finite- $SU(2)$  tower ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ), the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  equal to the affine- $E_8$  Kac marks ( $\sum = 30 = h(E_8)$ ), and it contains the 2, 3, 5 rotation-axis orders. This does *not* break B’s verdict — it is a *backward* certificate (it presupposes  $E_8$ ), so  $g_{\text{car}} = 5$  stays the P2 interface [O]; what it removes is the impression that  $(2, 3, 5)$  is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Still open, typed honestly: A (i)+(ii); Target D’s CP-phase residual (at  $\theta_{13} = 0$  the Dirac phase is undefined, so this needs a texture correction); Target E’s reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

**Second follow-up round [E]** ([verification/v87\_bulk\_uniqueness\_reduction.py], [verification/v88\_cp\_phase\_audit.py], [verification/v86\_nstar\_reheating.py])

**Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual.** Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems  $\simeq$  physical modular invariants of  $\text{Rep}(A)$ . For a *holomorphic* net  $\text{Rep}(A) = \text{Vect}$ , so the bulk pairing is *unique*. The contrast is machine-checked: the same- $c$  rival  $SO(16)_1$  (discriminant category  $\mathbb{Z}_2 \times \mathbb{Z}_2$ ,  $\mu$ -index 4) admits *six* modular invariants — diagonal,  $s \leftrightarrow c$  swap, both  $E_8$ -extension invariants  $|\chi_0 + \chi_{s/c}|^2$  (the lattice glue  $E_8 = D_8 \cup (D_8 + s)$ ,  $h_s=1$  bosonic), and two twisted pairings — so without holomorphy the  $c = 8$  bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem “seam–Calderón boundary net is holomorphic with  $c = 8$ ” ([C]/[O]; ledger GATE.METRIC.05). The theorem also has an equivalent, more tractable *index* form ([verification/v89\_carrier\_index\_lemma.py], ledger GATE.METRIC.06): by Kawahigashi–Longo–Müger,  $\mu_A = [B:A]^2 \mu_B$ , the carrier inclusion has Jones index  $[(E_8)_1 : (D_5)_1 \times (A_3)_1] = \sqrt{16} = 4 = |\mu_4|$  — the glue-group order *is* the inclusion index — and the extension is the  $\mu_4$  simple-current extension (all three glue sectors are  $h = 1$  currents;  $248 = 45 + 15 + 64 + 64 + 60$ ). Since an index-4 simple-current extension of the carrier has  $\mu = 16/4^2 = 1$ , **holomorphy follows from the index statement:** Target A  $\Leftrightarrow$  “the seam net contains the carrier net with Jones index 4 as its  $\mu_4$  simple-current extension” — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([verification/v92\_glue\_uniqueness.py], ledger GATE.METRIC.07): exhaustive classification of the carrier discriminant form  $(\mathbb{Z}_4 \times \mathbb{Z}_4, q = (5x^2 + 3y^2)/8)$  shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet  $\mathbb{Z}_2$ ; the same two objects as the



$E_8$ -extension invariants above), and exactly one halfway extension, whose induced form  $q = \{0, 0, 0, \frac{1}{2}\}$  is the  $D_8$  discriminant form: the  $SO(16)_1$  rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier ( $\mu=16$ )  $\rightarrow SO(16)_1$  ( $\mu=4$ )  $\rightarrow (E_8)_1$  ( $\mu=1$ , sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

**Target A, final form — the Simple-Current Extension Theorem [E]/[C]** ([verification/v154\_simple\_current\_theorem.py]). Assembled into one statement: the carrier net  $A = (D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L = \langle (1, 1) \rangle$  has  $[B : A] = |L| = 4$ ,  $c = 5 + 3 = 8$ ,  $\mu(B) = \mu(A)/|L|^2 = 16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$ . The *entire* adversarial residual of Target A is now the single seam-side identification “the RP seam–Calderón net *is*  $A$ ” — the algebra  $(B \cong (E_8)_1)$  is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core (“the kernel is quasi-free”) *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN =  $|k|$ , 16 free Majoranas, currents  $248 = 120 + 128$ , holomorphic character  $E_4/\eta^8 = j^{1/3}$  with 248 at level 1 (v156). *Final reduction* (v160–v165): the free-bulk premise is recast as a fixed-point theorem (quasi-free  $\Rightarrow$  all higher cumulants  $\kappa_{2n} = 0$ , v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap  $(2/3)^6$  (v161), whose value is forced by  $\mu_4$ -equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items —  $A_2$  net existence (an assembly of established net constructions, [C]) and  $R_1 = \text{GATE.QGEO}$  (“seam boundary =  $\mathbb{P}^1 \setminus \mu_4$ ”) — with the irreducible core  $\{\pi, v_{\text{geo}}\}$  a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. *The two structural residuals are then discharged to a theorem* (v175): the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  makes full-cone reflection positivity reduce *for every*  $m$  to the one-particle contraction (verified on the complete  $2^{16}$ -dim Fock space: contraction, fixed multiplicity  $2^8$ , sub-leading =  $(2/3)^6$ ), and  $A_2$  is an assembled and verified  $(E_8)_1$  certificate (character  $1 + 248q + 4124q^2 + \dots$ , embedding  $248 = 120 + 128$ ,  $E_8$  Cartan even unimodular). Both then need the *same* single input — the seam realisation QGEO.REALIZE.01 — which stays the one irreducible [O]; net existence and full-cone RP become [E]. *That single residual is then driven to bedrock* (v176–v181): it is assembled as one central theorem (v176), split into the marks + kernel obligations whose finite cores close (v177/v178), unified into one conformal-realisation premise (v179), and reduced — via uniformisation, Kerékjártó and the order-4 Möbius classification — to a seam-*isometry* premise (v180) and finally to the strictly weaker conformal-*symmetry* / deck premise QGEO.SYM.01 (v181): the carrier  $\mu_4$  clock is the conformal deck of the seam. Below that the residual is *definitional*, not a missing theorem — the single irreducible structural residual of the whole theory. It is since driven to its sharpest, *non-circular* form (v194–v198): the raw seam DtN is RP-canonical (Osterwalder–Schrader on the quasi-free state), and the bedrock is cracked to the *state-invariance*  $\omega \circ \rho = \omega$  — the DtN principal symbol commutes *exactly* with the clock, and Tomita–Takesaki gives  $[\rho, \Lambda_\Sigma] = 0$  with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity — the maximally-operational form of the same postulate, still [O]. *The geometric and conditional cores are now Lean-formalised* (AUDIT: PASS): the UNIFORM normal form ( $z \mapsto iz$ ,  $\sigma\rho\sigma = \rho^{-1}$ , orbit  $\rightarrow \mu_4$ ; FORM.QGEO.02) and the implication mark-local  $\Rightarrow \omega \circ \rho = \omega$  (FORM.QGEO.01) — so the deduction below the premise is machine-proved, while QGEO.SYM.01 itself stays the one [O] postulate.

**Target D — the CP-phase residual quantified [E] (not closed).** With the frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$ : pull vs  $\gamma_{\text{PDG}} = 65.7^\circ \pm 3.0^\circ$  is  $+0.98\sigma$  (survives). Audit, *not* promoted: the data central value sits  $0.07^\circ$  from the alternative coefficient reading  $\pi/3 + 2\lambda_C^2$  ( $|\mathbb{Z}_2|$  instead of  $N_{\text{fam}}$ ) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (REG.FREEZE.01). Decision threshold: the readings differ by  $\lambda_C^2 = 2.88^\circ$ , i.e.  $3\sigma$ -distinguishable once  $\sigma_\gamma \leq 0.96^\circ$ . The  $J$ -inversion is flagged magnitude-contaminated (source-scale  $s_{23}$  vs  $M_Z$ ). Ledger FLAV.CP.01.

**Target E — one flagged scale computed [C].** The reheating temperature is no longer a silent input:  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  [E] + standard physics give  $T_{\text{reh}} = 9.6 \times 10^9$  GeV and  $N_*(0.05/\text{Mpc}) = 51.4$  [C] (ledger COSMO.NSTAR.01); honestly recorded: this slow Higgs-channel point is  $A_s$ -*disfavoured* ( $-11.4\sigma$ ; the measured  $A_s$  requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band [50, 60] stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was merged (A), one was quantified (D), one input was computed (E).**

**Target D — the CP residual geometrically *located* (not closed) [E]/[C]**

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. **(i) A channel typing** ([`verification/v227_degree_exponent_channel_split.py`]). The  $E_8$  split  $248 = 120 + 128$  is a magnitude/phase split: the exponents sum to  $120 = |R^+(E_8)|$  (the *magnitude* channel, where the (ratios, product)  $\Rightarrow v_{\text{geo}}$  bijection lives), the fundamental degrees sum to  $128 = \text{rank } E_8 \cdot \dim S^+$  (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong “ $S^-$  is dark matter” reading). **(ii) A hexagonal modulus** ([`verification/v220_cp_hexagonal_modulus.py`]). The seam deck is the *square* modulus ( $j=1728, \mathbb{Z}/4$  clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus  $j=0$  ( $\tau=\rho=e^{i\pi/3}, \mathbb{Z}/6$ , CM by  $\mathbb{Z}[\omega]$ ). The frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$  has leading term  $\pi/3 = \arg \rho$ , the hexagonal CM phase; CP lives in the  $\mathbb{Z}/6$  phase *fiber* over the  $\mathbb{Z}/4$  seam, not in the deck. **(iii) An orientation** ([`verification/v225_dual_normal_frame.py`]). The dual normal frame  $d = a^\top R^{-1} = -\frac{1}{2}(1, 1, -2)$  (Nariai/horizon normal) and  $n = (5, -9, 6)$  (first-generation selector,  $n^\top R = (8, 0, 0)$ ) has oriented volume  $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$ , and rotating by the hexagonal unit gives  $\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3) \neq 0$  — CP sits precisely in the *orientation* of the normal frame, exactly where the magnitude bijection fails. All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry  $\delta$  is untouched). **(iv) One hexagonal unit, two sheets** ([`verification/v231_cp_mu6_phases.py`]). The reduction is sharper still: *both* CP phases are  $\mu_6$  powers of the single hexagonal CM unit  $\rho = e^{i\pi/3} - \delta_{\text{CKM}}^{\text{lead}} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the neutrino phase lattice) — with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  is the quark phase times the  $\mathbb{Z}_2$  sheet ( $\rho^3 = -1$ ); the frame orientations of (iii) are correspondingly sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). So Target D’s CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual  $3\lambda_C^2$  (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. **(v) The universal triality phase** ([`verification/v233_cp_triality_phase.py`]). In the canonical  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation,  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight); since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of  $\mu_6$  power: the power choice of (iv) is removed, and  $\omega$  is the  $\mu_3$  factor of the order-30 (2, 3, 5) Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C].

**Global look-elsewhere closure [E] ([`verification/v100_numerology_null_mc.py`])**

The per-target look-elsewhere traps recorded above (the  $\pi/3 + 2\lambda_C^2$  alternative in Target D, the  $\ln(46080)$  scale reading destroyed in [`verification/v99_koide_flow_time.py`]) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability  $\prod_i p_i = 10^{-25.8}$  over the 13 scored observables; all 94 500  $F_{U(1)}$  variants solved, exactly one hits CODATA ( $p_\alpha \leq 1.1 \cdot 10^{-5}$ ); total  $\leq 10^{-30.7}$ . The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT’s own hit count  $13 \rightarrow 9/6$ ; data shuffling  $\rightarrow 1.2$ ) prove it has power, and the census is stable under budget/window variation ( $< 2$  orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (tfpt\_3) and the module docstring.

**External-review residual map [E]/[C] ([`verification/v182_reviewer_residual_map.py`])**

A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: **(A)** the mapping  $2D \rightarrow 4D$  (“why does the dynamics read the masses as the Plücker norm 11?”), **(B)** the abstractness of UV gravity, **(C)** the meta-look-elsewhere of the grammar choice, and **(D)** the Koide Möbius flow (non-integer flow time  $t \sim 2.84$ , missing QFT generator). None adds an *independent* open gap. **A** [E]/[C]: the readout integer is exact ( $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}}$ , the QBL boundary count, v174) and the  $2D \rightarrow 4D$  step is the holographic reduction, so  $A = \text{QGE0.SYM.01} + F_{\text{transfer}}$ . **B** [E]: “gravity is the geometric readout of the boundary” *is* QGE0.SYM.01. **D** [E]/[C]:  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  is the Möbius group and the cusp flow

lives on the deck  $\mathbb{P} \setminus \mu_4$ , so the Möbius *nature* is forced by the geometry; the non-integer  $t \sim 2.84$  only locates the physical point and the missing generator *is*  $F_{\text{transfer}}$ , so  $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (v84), minimal and over-determined (the anchor  $a = (1, 1, 2)$  lands in  $\geq 6$  mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay **[O]**/**[C]**.